



OMNISENSUS MEDICAL

AI-Powered Clinical
Intelligence Platform

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OUTLINE

- 1. Problem Statement — Healthcare & Medical Analytics (Domain 5)**
2. Proposed System / Solution
3. System Development Approach (Technology Used)
4. Algorithm & ML Model
5. Deployment Architecture
6. Result & Output (Screenshots)
7. Conclusion
8. Future Scope
9. References

PROBLEM STATEMENT

Problem Domain 5: Healthcare & Medical Analytics

① Rising Disease Prevalence

Diabetes, heart disease, and CKD are increasing due to lifestyle changes and ageing populations — yet hospitals lack any automated real-time early-warning system to identify at-risk patients proactively.

② Delayed Diagnosis

Manual analysis of patient biomarkers (glucose, HbA1c, eGFR, blood pressure, LDL) across paper records and siloed EMR systems causes critical diagnostic delays and increases patient risk.

③ Healthcare Resource Constraints

Hospitals struggle to optimise bed allocation, staff scheduling, and equipment usage without real-time patient acuity data — resources are allocated reactively, not intelligently.

④ Data Complexity & Fragmentation

Patient records are heterogeneous, unstructured, and spread across multiple disconnected sources — making it impossible to build a unified view of multi-domain health status.

⑤ Predictive Uncertainty

Anticipating patient outcomes, readmission risks, and disease progression remains challenging — no automated scoring system exists to combine multi-domain risk into a single actionable score.

PROPOSED SYSTEM / SOLUTION

OmniSensus Medical

A full-stack, AI-powered Clinical Decision Support System that automates multi-domain disease risk stratification in real time.

Three ML Models

Cardiovascular, Diabetes (Metabolic), Kidney (CKD) — each a calibrated Random Forest classifier producing disease risk probabilities.

Composite Health Score (0–100)

Weighted formula combining the three domain risks into a single patient wellness score, classified as Stable / Borderline / Critical.

Three Role Dashboards

System Admin, Doctor, and Patient — each with purpose-built workflows, data access, and dedicated UI.

91.3%

Cardiovascular AUC

82.1%

Diabetes AUC

100%

Kidney (CKD) AUC

27 Tables | 8 Views | 12 Triggers
60+ REST API Endpoints | 3 User Roles

SYSTEM DEVELOPMENT APPROACH

FRONTEND

Next.js 14 (App Router)
React 18 — SSR + CSR
Chart.js 4.4 (Radar, Line, Bar)
27-file modular CSS design system
Role-specific layouts (Admin, Doctor, Patient)

BACKEND API

FastAPI (Python 3.11)
JWT Auth + Refresh Tokens
60+ REST endpoints
asynpg + SQLAlchemy ORM
Role-based middleware (RBAC)

ML SERVICE

Flask microservice
CalibratedClassifierCV (RF)
SHAP explainability engine
ARIMA(1,1,0) trend forecasting
Qwen 0.5B LLM (PharmaBot)

DATABASE: PostgreSQL 16 (Neon Serverless) — 27 Tables | 8 Views | 12 Triggers

DEPLOYMENT: Render (FastAPI + Flask) | Netlify (Next.js) | GitHub CI/CD

Tools: VS Code | Postman | pgAdmin | GitHub | Neon Console | Render Dashboard

ALGORITHM & ML MODEL

ML Pipeline — Per Disease Domain:

1. Data → Column Normalise → Target Binarise
2. Median Imputation for missing values (robust to clinical outliers)
3. StandardScaler (Z-score normalisation)
4. SMOTE oversampling + `class_weight="balanced"` (dual imbalance handling)
5. CalibratedClassifierCV (RandomForest, method="sigmoid") — Platt Scaling
6. Decision threshold = 0.35 (adaptive, context-aware; floor = 0.20)

Composite Health Score Formula:

Risk = (Heart × 0.40) + (Diabetes × 0.35) + (Kidney × 0.25)

Health Score = (1 – Risk) × 100 | capped at 95 for age ≥ 70

HEART DISEASE

91.3%

AUC — ROC Score
UCI Cleveland (303 rows, 13 features)

DIABETES

82.1%

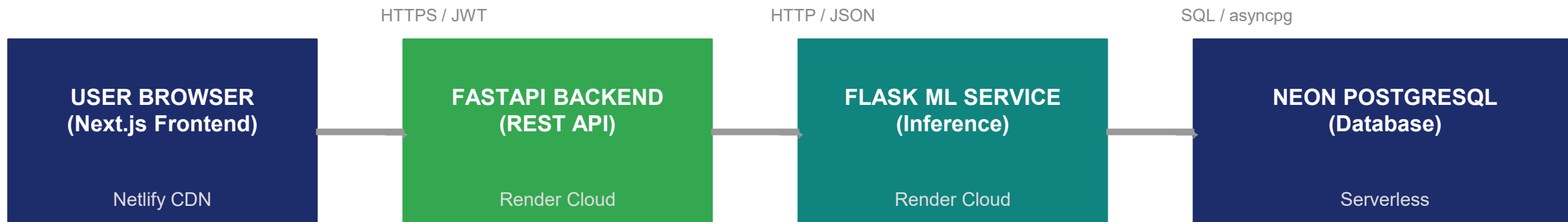
AUC — ROC Score
Pima Indians (768 rows, 8 features)

KIDNEY (CKD)

100%

AUC — ROC Score
UCI CKD (400 rows, 15 features)

DEPLOYMENT ARCHITECTURE



Key Architectural Decisions:

Microservices separation — ML workload (SHAP, PDF, inference) runs independently so it never blocks web API request processing.

JWT Bearer authentication with 30-min access token expiry and secure refresh token rotation — enforced at every FastAPI endpoint via role-based middleware.

Neon Serverless PostgreSQL — auto-scaling connection pooling, SSL-enforced, zero-infrastructure management. 27 tables | 8 views | 12 triggers.

Render free-tier deployment — zero-cost cloud hosting with a clear upgrade path to paid tiers for production-scale usage.

RESULT & OUTPUT

ML MODEL PERFORMANCE

Heart Disease (CVD)

UCI Cleveland — 303 samples, 13 features

91.3%

AUC-ROC

Precision: 88% Recall: 90% F1-Score: 89%

Diabetes (Metabolic)

Pima Indians — 768 samples, 8 features

82.1%

AUC-ROC

Precision: 79% Recall: 81% F1-Score: 80%

Kidney Disease (CKD)

UCI CKD — 400 samples, 15 features

100%

AUC-ROC

Precision: 100% Recall: 100% F1-Score: 100%

SYSTEM OUTPUT — WHAT THE PLATFORM PRODUCES



Composite Health Score (0–100)

Risk formula:

(Heart×0.40)+(Diabetes×0.35)+(Kidney×0.25). Classified as Stable / Borderline / Critical.



SHAP AI Insights Per Diagnostic Run

Top-10 biomarker contributors — directional, patient-specific, stored in ai_insights DB table.



Automated PDF Diagnostic Report

Auto-generated per run: biomarker table, domain scores, clinical flags, SHAP rankings.



Clinical Flags & Triage Urgency

3-tier flags (Critical/Borderline/Info) + department routing (Cardiology, Nephrology, etc.).



Trend Forecast — ARIMA(1,1,0)

Next-visit health score prediction from longitudinal history. Needs minimum 4 visits.



Readmission Risk Score (0–100)

5-component model: health score + trend + flags + biomarker danger zones + context.

OUTPUT

MAIN LOGIN PAGE

FOR ADMIN : ->

USERNAME: MS.Elevate

PASSWORD: MSElevate@2026!

FOR DOCTOR : ->

USERNAME : Dr.Sharma

PASSWORD: MedPass24

FOR PATIENTS : ->

USERNAME: P-00001

PASSWORD: Health2026



OmniSensus

Welcome Back

Please sign in to access your dashboard.

Institutional Username

 MS Elevate

Password

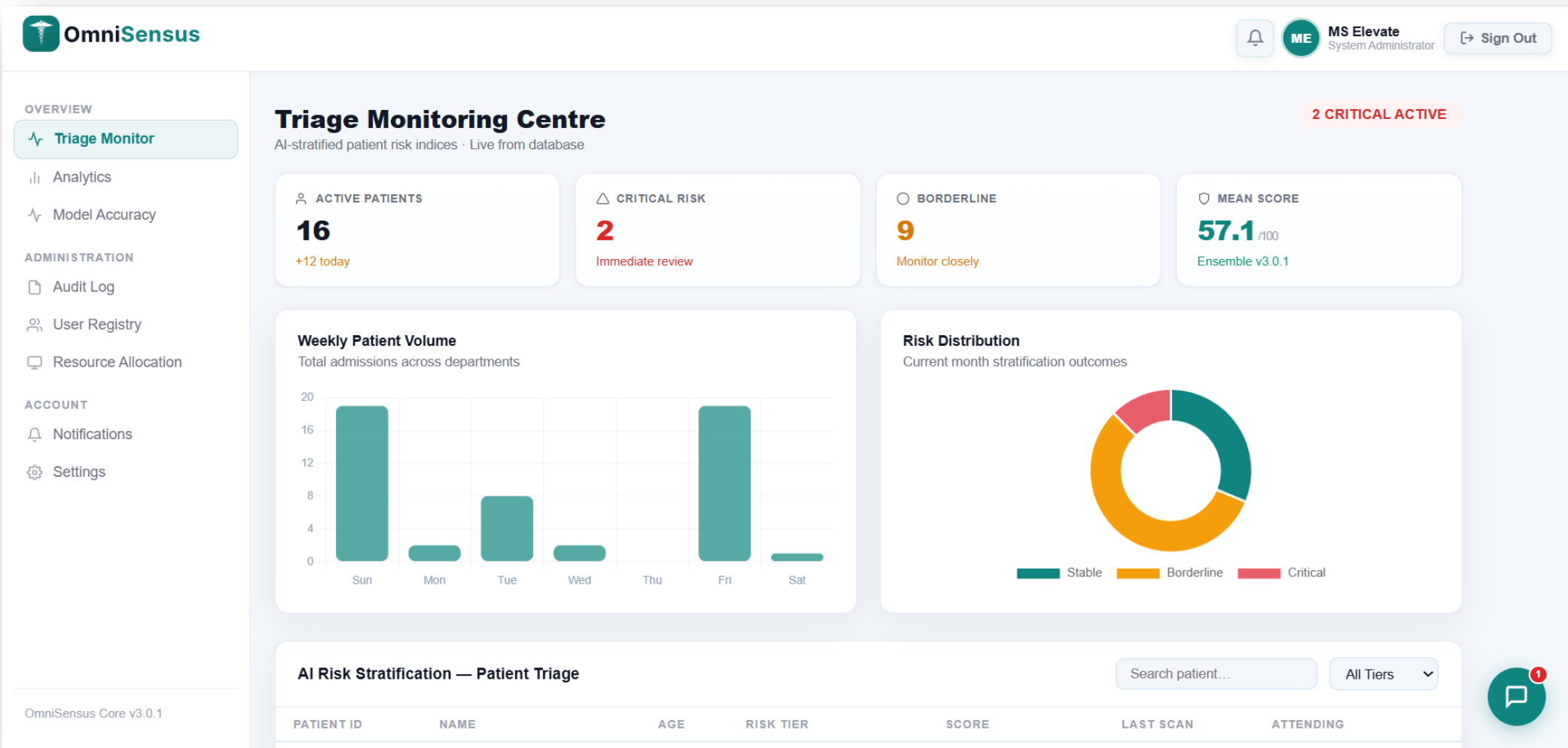
 MSElevate@2026! 

 Sign In

Unauthorised access is strictly prohibited and monitored.

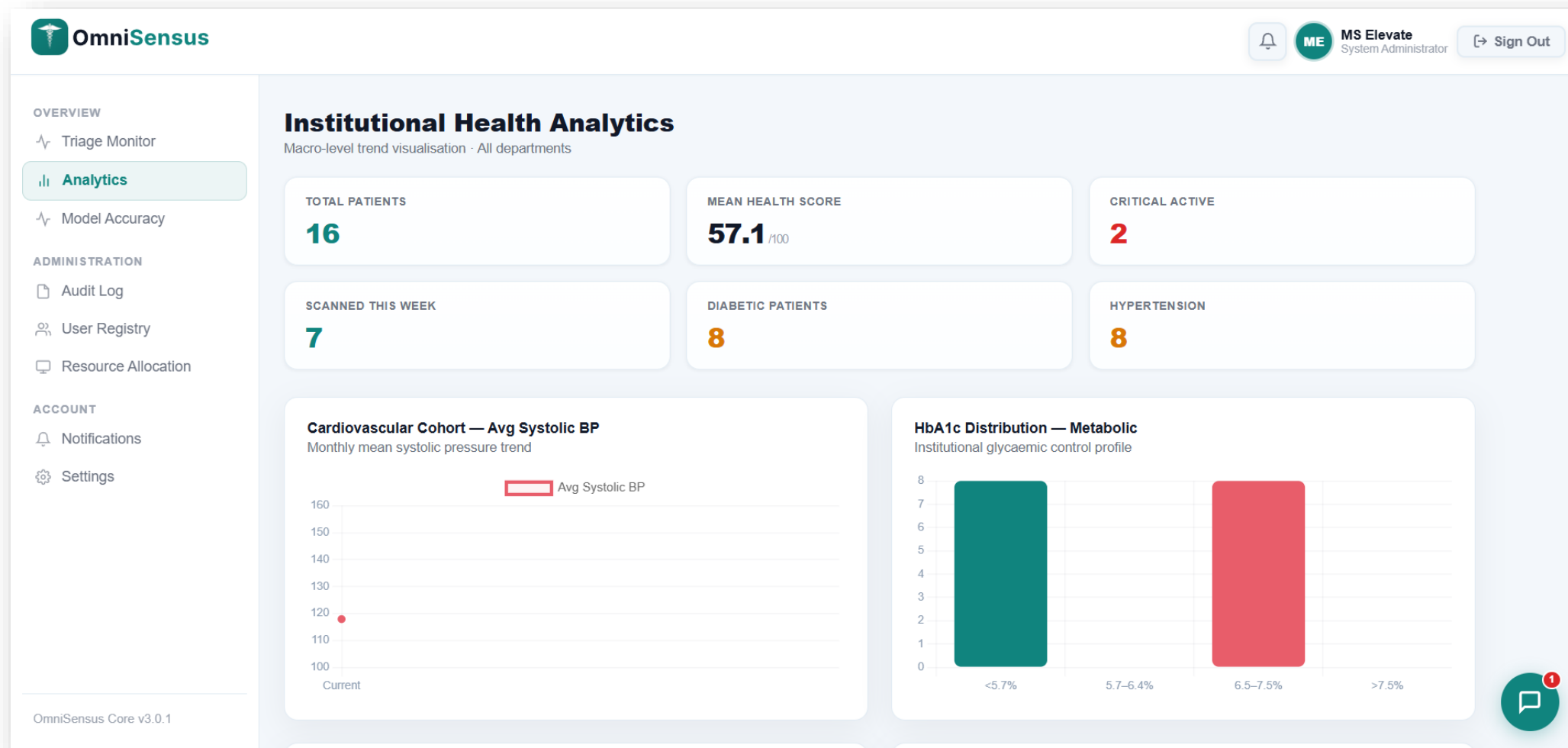
OUTPUT : ADMIN MODULE

TRIAGE MONITOR PAGE (DASHBOARD)



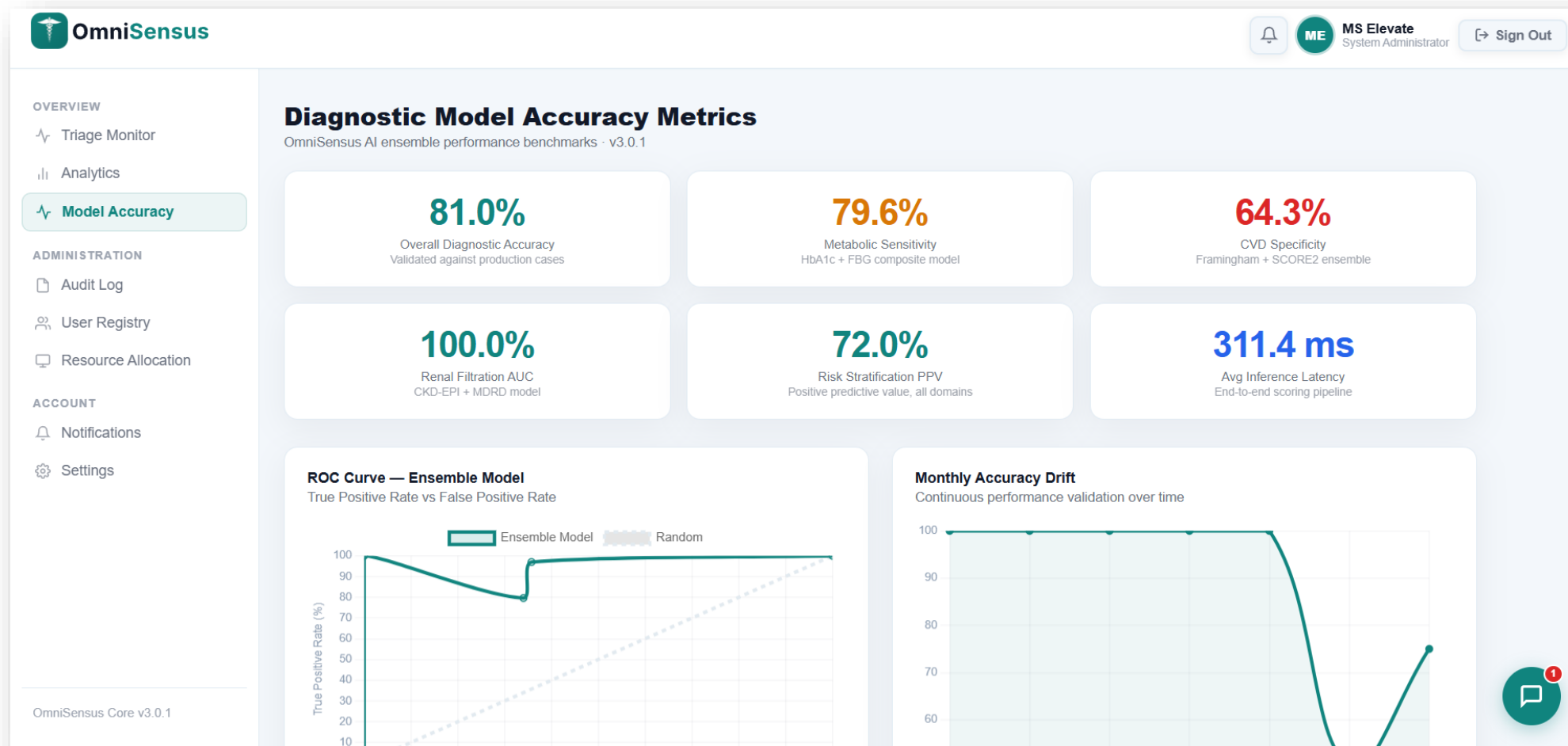
OUTPUT : ADMIN MODULE

ADMIN ANALYTICS PAGE



OUTPUT : ADMIN MODULE

MODEL ACCURACY PAGE



OUTPUT : ADMIN MODULE

AUDIT LOG PAGE


OmniSensus



MS Elevate
 System Administrator
 [Sign Out](#)

OVERVIEW
 Triage Monitor
 Analytics
 Model Accuracy

ADMINISTRATION
Audit Log
 User Registry
 Resource Allocation

ACCOUNT
 Notifications
 Settings

OmniSensus Core v3.0.1

System Audit Log

Chronological stream of database transactions and security events

Filter by action...
 Search

Transaction & Security Event Stream
195 TOTAL EVENTS

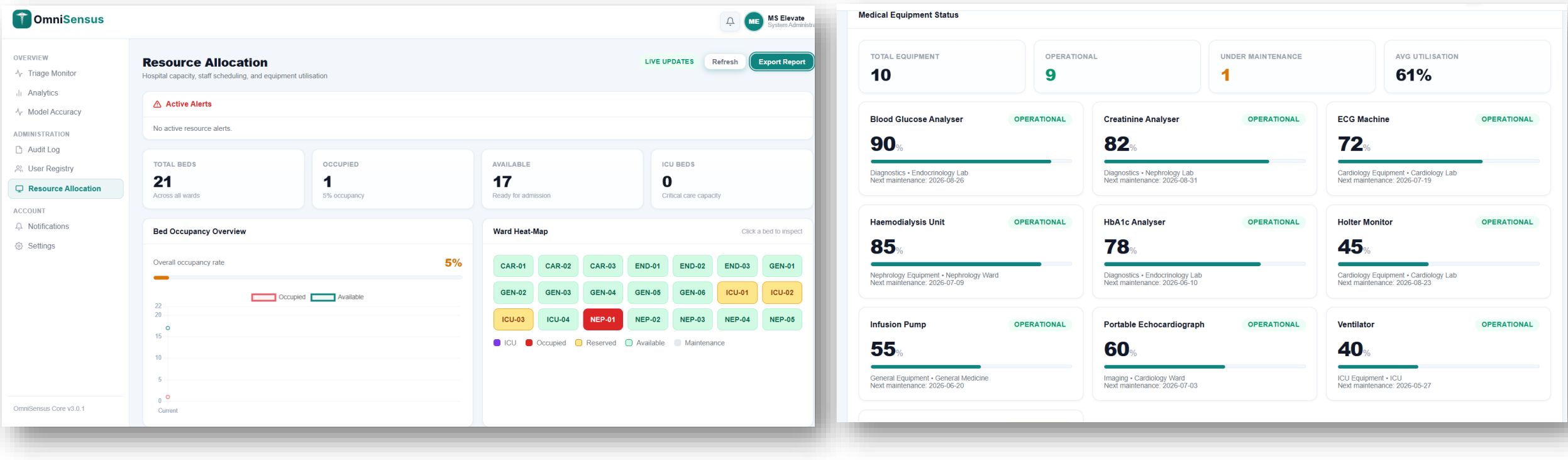
07:06:17 PM	✓	LOGIN · Auth · User: MS.Elevate	TXN-D2EA6834	DATA
06:46:25 PM	✓	LOGIN · Auth · User: MS.Elevate	TXN-596BCDCF	DATA
05:40:06 PM	✓	LOGIN · Auth · User: Dr.Sharma	TXN-E934C34F	DATA
05:23:52 PM	✓	LOGIN · Auth · User: Parth	TXN-51E6E668	DATA
03:55:17 PM	✓	LOGIN · Auth · User: P-00001	TXN-589426B9	DATA
03:39:10 PM	✓	LOGIN · Auth · User: Dr.Sharma	TXN-DCC656D1	DATA
03:08:33 PM	✓	LOGIN · Auth · User: Parth	TXN-EC7DF7D7	DATA
02:52:02 PM	✓	LOGIN · Auth · User: Parth	TXN-EFAB4F0A	DATA
02:23:37 PM	✓	LOGIN · Auth · User: Dr.Sharma	TXN-40AF2D4C	DATA
02:23:14 PM	✓	LOGIN · Auth · User: Parth	TXN-93F429DF	DATA





OUTPUT : ADMIN MODULE

RESOURCE ALLOCATION PAGE



OUTPUT: DOCTOR MODULE

DIAGNOSTIC WIZARD STEP-1 (VITALS INPUT)



OmniSensus



DR

Dr. Rajesh Sharma

Clinical Practitioner

Sign Out

CLINICAL

Diagnostic Engine

AI Lab

My Patients

Reports

Schedule

ACCOUNT

Notifications

Settings

AI-Powered Risk Stratification for Clinical Insights

Search symptoms, biomarkers, or lab tests...

Run Scan

Fatigue & Weakness

Elevated Glucose

Dyspnea on Exertion

Polyuria / Polydipsia

Hematuria

CARDIAC RHYTHM

127 bpm

Elevated

ARTERIAL PRESSURE

150/76 mmHg

Hypertensive

GLYCEMIC INDEX

89.3 mg/dL

Normal

COMPOSITE RISK INDEX

62%

Elevated Risk

Diagnostic Engine — 3-Step Protocol

Vitals

Biomarkers

History

Nisha Agarwal · c3198711

HEART RATE (BPM)

127

SYSTOLIC BP (MMHG)

150

DIASTOLIC BP (MMHG)

76

RESPIRATORY RATE (/MIN)

0

BODY TEMP (°C)

0.0

SPO2 (%)

82

Health Status Card

38

Health Score

Elevated Risk

Immediate clinical review required

Cardiovascular

Metabolic

Renal

Low

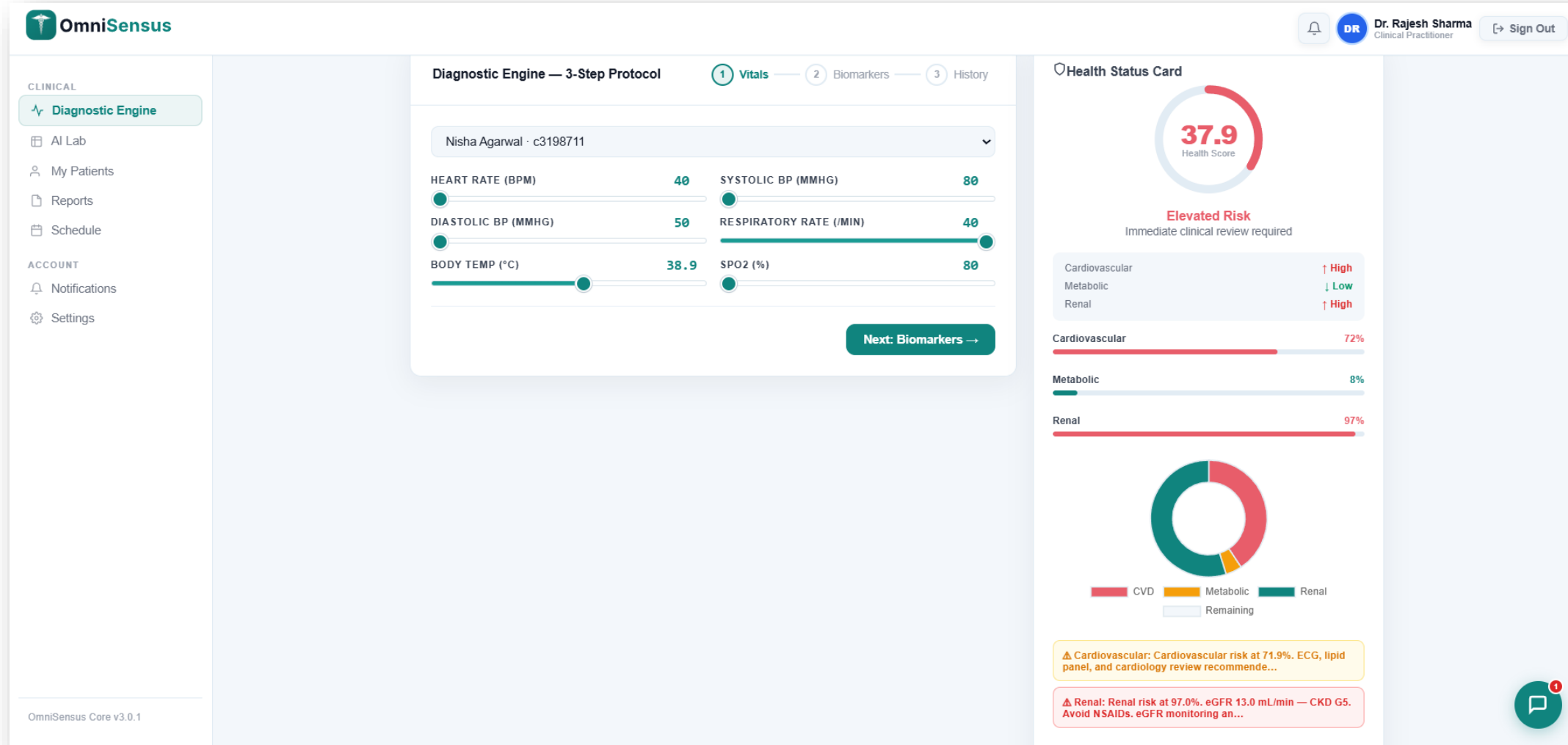
Low

Low

OmniSensus Core v3.0.1

OUTPUT: DOCTOR MODULE

HEALTH SCORE RING + DOMAIN RISK CHART



OUTPUT: DOCTOR MODULE

PHARMABOT CHAT



DR
Dr. Rajesh Sharma
Clinical Practitioner

 Sign Out

AI-Powered Risk Stratification for Clinical Insights

Run Scan

Fatigue & Weakness

Elevated Glucose

Dyspnea on Exertion

Polyuria / Polydipsia

Hematuria

 CARDIAC RHYTHM
40 bpm
↓ Normal Sinus

 ARTERIAL PRESSURE
80/ 50 mmHg
↓ Optimal

 GLYCEMIC INDEX
50 mg/dL
— Normal

 COMPOSITE RISK
62.1 %
Elevated Risk

Diagnostic Engine — 3-Step Protocol

✓ Vitals

✓ Biomarkers

3 History

Nisha Agarwal · c3198711

AGE (YEARS)

38

SEX AT BIRTH

Female

BMI

23.7

SMOKING STATUS

Never

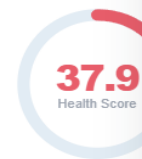
FAMILIAL CVD HISTORY

No

METABOLIC PATHOGENESIS

No

 Health Status Card



37.9
Health Score

Elevated Risk
Immediate clinical review

Cardiovascular

Metabolic

Renal

Cardiovascular

0%

Metabolic

0%


OmniSensus Pharma-Bot
AI Medical Assistant - Online

CLR X

medications, risk factors, diagnostic interpretations, and medical queries. How can I help?

How many patients do I have right now?

02:27 PM

You currently have 7 patients in your panel.

02:27 PM

How many patients do I have right now?

How to run diagnostic for a patient?

Where can I download the latest report?

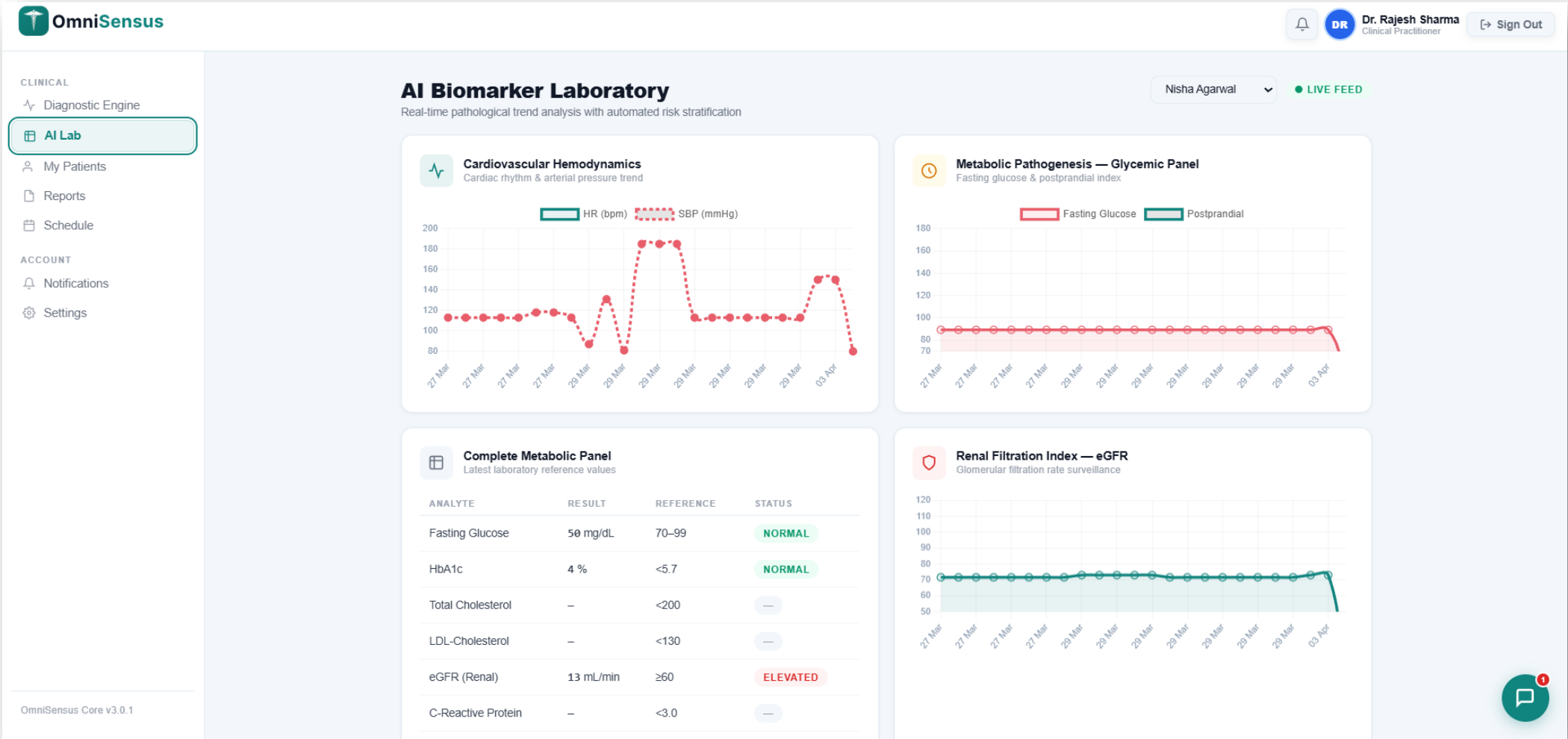
Why is my access not working?





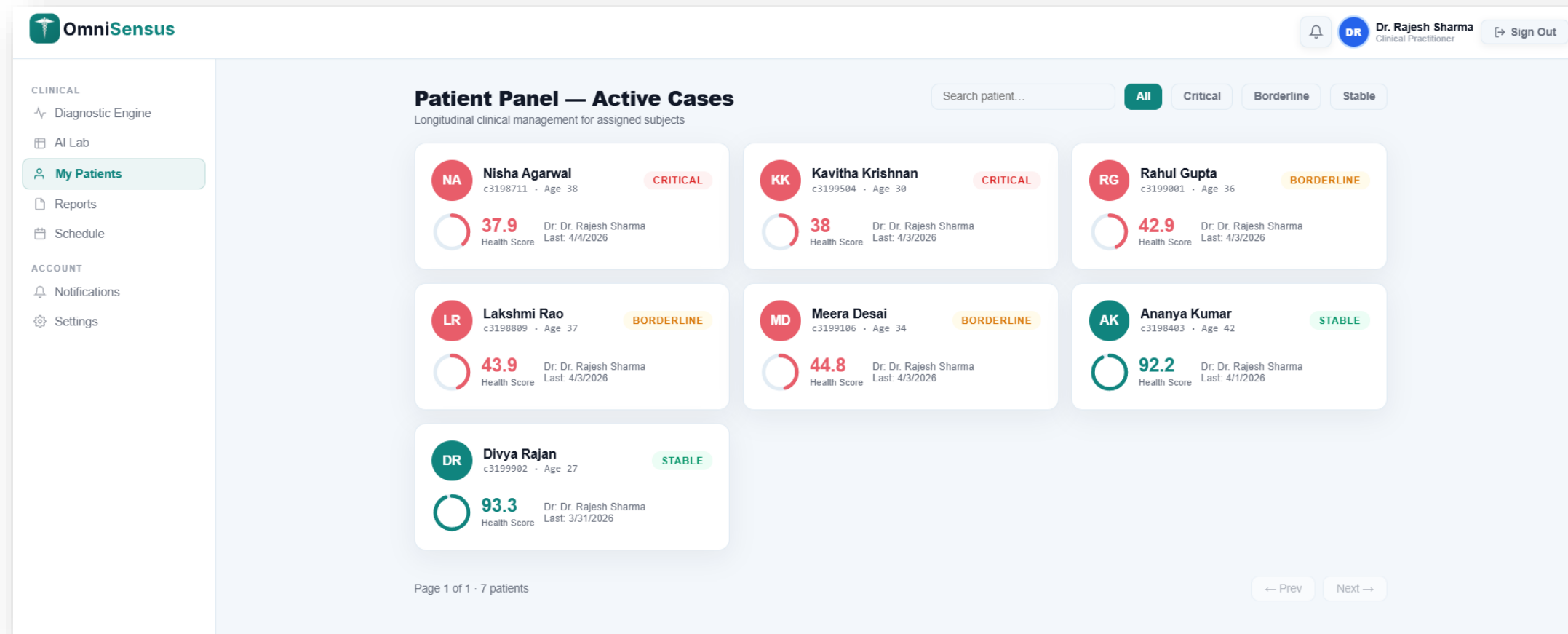
OUTPUT: DOCTOR MODULE

AI LAB DASHBOARD



OUTPUT: DOCTOR MODULE

MY PATIENTS DASHBOARD



OUTPUT: DOCTOR MODULE

DOCTOR REPORTS PAGE

OmniSensus

CLINICAL

Diagnostic Engine

AI Lab

My Patients

Reports

Schedule

ACCOUNT

Notifications

Settings

Clinical Report Archive

AI-generated diagnostic reports — view and download

Nisha Agarwal

full_diagnostic

03 April 2026 · Score: 38/100 · Acc: OSM-ACC-000030

CRITICAL

View

Download

full_diagnostic

03 April 2026 · Score: 38/100 · Acc: OSM-ACC-000029

CRITICAL

View

Download

full_diagnostic

30 March 2026 · Score: 92.2/100 · Acc: OSM-ACC-000024

STABLE

View

Download

full_diagnostic

30 March 2026 · Score: 92.2/100 · Acc: OSM-ACC-000023

STABLE

View

Download

full_diagnostic

29 March 2026 · Score: 44.2/100 · Acc: OSM-ACC-000021

BORDERLINE

View

Download

full_diagnostic

29 March 2026 · Score: 85.6/100 · Acc: OSM-ACC-000020

STABLE

View

Download

full_diagnostic

29 March 2026 · Score: 89.6/100 · Acc: OSM-ACC-000019

STABLE

View

Download

full_diagnostic

28 March 2026 · Score: 92.2/100 · Acc: OSM-ACC-000018

STABLE

View

Download

OmniSensus Core v3.0.1

AFTER CLICK ON VIEW BUTTON

Report Preview

Review all report details below. Download protected PDF report.

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Date: 4/3/2026

Acc: OSM-ACC-000030

HEALTH SCORE

38

Clinical Visit

No additional notes.

CRITICAL

KEY BIOMARKERS

PARAMETER	VALUE
Fasting Glucose	89.3 mg/dL
HbA1c	5.5%
eGFR	73 mL/min
Blood Pressure	150/76 mmHg

Close

Download Secured PDF

OUTPUT: DOCTOR MODULE

DOWNLOADED REPORTS

Report Preview

Review all report details below. Download protected PDF report.

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Date: 4/3/2026

Acc: OSM-ACC-000030

HEALTH SCORE

Clinical Visit

38

No additional notes.

CRITICAL

KEY BIOMARKERS

PARAMETER	VALUE
Fasting Glucose	89.3 mg/dL
HbA1c	5.5%
eGFR	73 mL/min
Blood Pressure	150/76 mmHg

Close

Download Secured PDF

AFTER CLICK ON DOWNLOAD

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Report ID: OSM-ACC-000030

Generated: 03 Apr 2026

PATIENT

AGE / SEX

BLOOD GROUP

PHYSICIAN

Nisha Agarwal

38 / F

A+

Dr. Rajesh Sharma

Composite Health Score

38.0

Critical

Above 65 = Stable | 40-65 = Borderline | Below 40 = Critical

28.1

Cardiovascular

91.9

Metabolic

3.0

Renal

Biomarker Results

Biomarker	Value	Unit	Status
Blood Glucose	89.3	mg/dL	Normal
HbA1c	5.5	%	Normal
Systolic BP	150.0	mmHg	Normal
Diastolic BP	76.0	mmHg	Normal
eGFR	73.0	mL/min	Normal

Disease-Specific Risk Probabilities (ML Model)

Disease Domain	Risk (%)	Threshold	Status
Cardiovascular Disease	28.1%	35%	Low Risk
Diabetes / Pre-Diabetes	91.9%	35%	High Risk
Chronic Kidney Disease	3.0%	35%	Low Risk

Recommended Referral

Clinical follow-up advised based on current risk profile.

Electronically generated by OmniSensus AI Engine v3.0.1 | Attending Physician: Dr. Rajesh Sharma | 04 Apr 2026

OUTPUT: DOCTOR MODULE

MY SCHEDULE PAGE



OmniSensus

CLINICAL

Diagnostic Engine

AI Lab

My Patients

Reports

Schedule

ACCOUNT

Notifications

Settings



DR

Dr. Rajesh Sharma

Clinical Practitioner



Sign Out

My Schedule

Upcoming and past appointments

+ New Appointment

Wednesday, 01 April 2026

10:30 AM

30min

Ananya Kumar

annual physical

Scheduled follow-up. Review latest diagnostics.

CONFIRMED

Done

Friday, 03 April 2026

11:15 AM

30min

Kavitha Krishnan

annual physical

Scheduled follow-up. Review latest diagnostics.

BOOKED

Confirm

Friday, 10 April 2026

10:00 AM

30min

Divya Rajan

annual physical

Scheduled follow-up. Review latest diagnostics.

BOOKED

Confirm

Sunday, 12 April 2026

02:15 PM

30min

Rahul Gupta

annual physical

Scheduled follow-up. Review latest diagnostics.

BOOKED

Confirm

Wednesday, 22 April 2026

11:00 AM

30min

Nisha Agarwal

annual physical

Scheduled follow-up. Review latest diagnostics.

BOOKED

Confirm

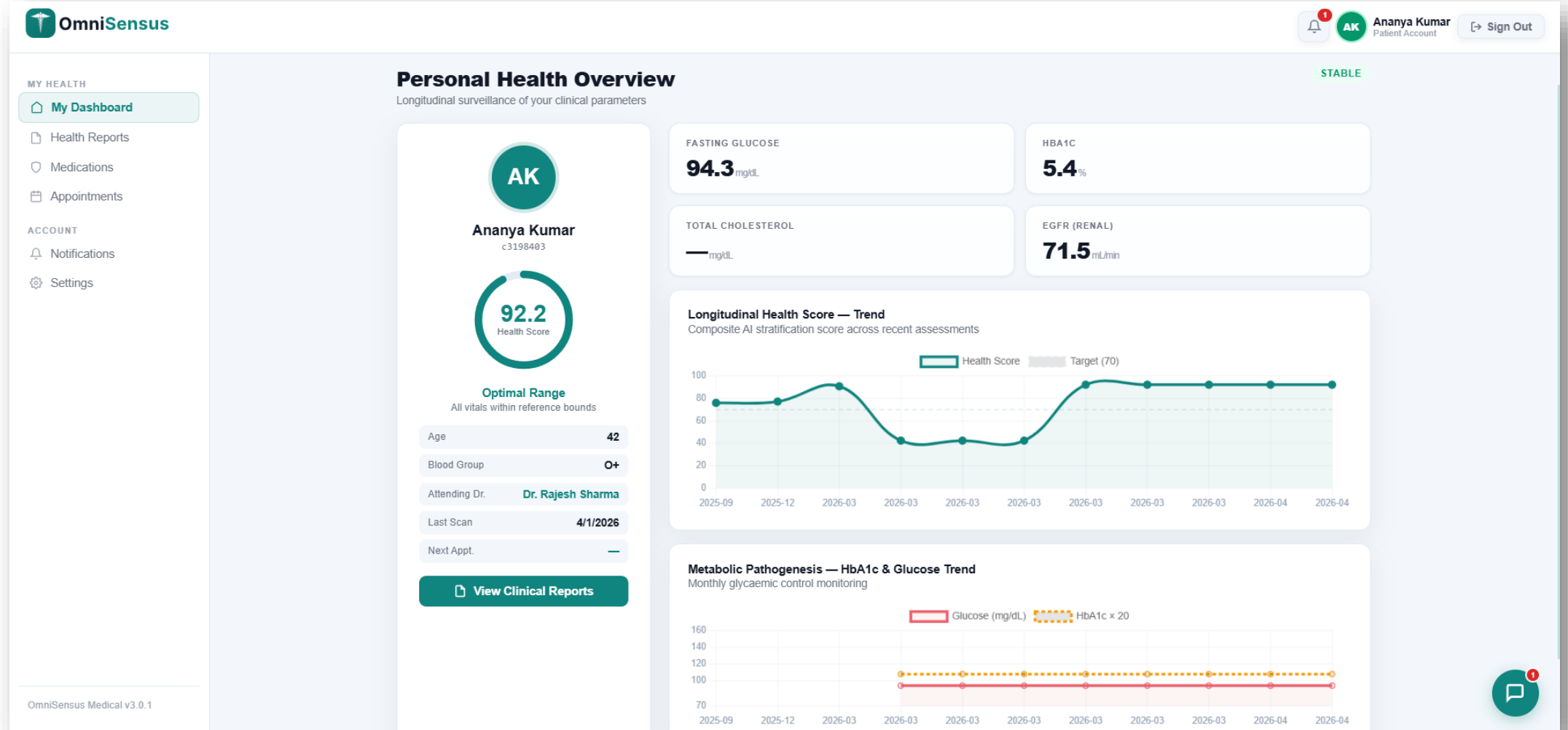
Friday, 24 April 2026

OmniSensus Core v3.0.1



OUTPUT: PATIENT MODULE

PATIENT DASHBOARD



OUTPUT: PATIENT MODULE

APPOINTMENT PAGE

 OmniSensus

MY HEALTH

My Dashboard

Health Reports

Medications

Appointments

ACCOUNT

Notifications

Settings

My Appointments

Live schedule from the hospital system

Upcoming (0)

No upcoming appointments.

History (3)

follow up

12/14/2025, 12:00:00 PM

Duration: 30 min · Doctor: Dr. Rajesh Sharma

Follow-up visit completed.

COMPLETED

follow up

2/13/2026, 2:00:00 PM

Duration: 30 min · Doctor: Dr. Rajesh Sharma

Follow-up visit completed.

COMPLETED

annual physical

4/1/2026, 10:30:00 AM

Duration: 30 min · Doctor: Dr. Rajesh Sharma

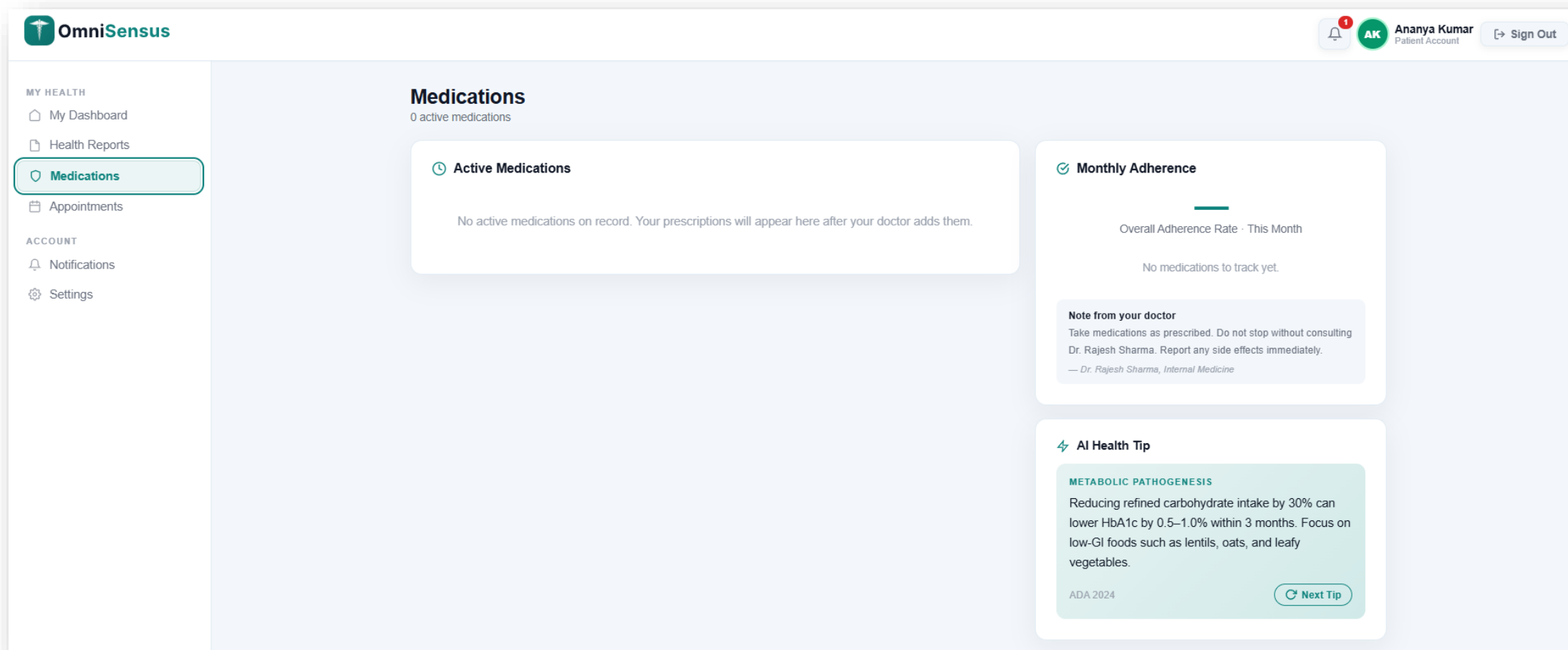
Scheduled follow-up. Review latest diagnostics.

CONFIRMED

OmniSensus Medical v3.0.1

OUTPUT: PATIENT MODULE

MEDICATION PAGE



OUTPUT: PATIENT MODULE

HEALTH REPORT PAGE



MY HEALTH

My Dashboard

Health Reports

Medications

Appointments

ACCOUNT

Notifications

Settings

full_diagnostic

16 March 2026 · Score: 90.8/100 · Acc: OSM-ACC-000001

STABLE

View

Download

Report Preview

Review all report details below. Download protected PDF report.

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Date: 3/16/2026

Acc: OSM-ACC-000001

HEALTH SCORE

90.8

STABLE

No additional notes.

KEY BIOMARKERS

PARAMETER	VALUE
Fasting Glucose	94.3 mg/dL
HbA1c	5.4%
eGFR	71.5 mL/min
Blood Pressure	115/77 mmHg

Close

Download

AFTER CLICK ON VIEW

OUTPUT: PATIENT MODULE

HEALTH REPORT DOWNLOADED

Report Preview

Review all report details below. Download protected PDF report.

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Date: 3/16/2026

Acc: OSM-ACC-000001

HEALTH SCORE

full_diagnostic

No additional notes.

90.8

STABLE

KEY BIOMARKERS

PARAMETER	VALUE
Fasting Glucose	94.3 mg/dL
HbA1c	5.4%
eGFR	71.5 mL/min
Blood Pressure	115/77 mmHg

Close

Download

AFTER CLICK ON DOWNLOAD

OmniSensus Medical

AI-Powered Clinical Diagnostic Report

Report ID: OSM-ACC-000001

Generated: 16 Mar 2026

PATIENT

AGE / SEX

BLOOD GROUP

PHYSICIAN

Ananya Kumar

42 / F

O+

Dr. Rajesh Sharma

c3198403-0800-4162-9246-8af...

Composite Health Score

90.8

Stable

Above 65 = Stable | 40-65 = Borderline | Below 40 = Critical

84.9

Cardiovascular

86.8

Metabolic

95.6

Renal

Biomarker Results

Biomarker	Value	Unit	Status
Blood Glucose	94.3	mg/dL	Normal
HbA1c	5.4	%	Normal
Systolic BP	115.0	mmHg	Normal
Diastolic BP	77.0	mmHg	Normal
eGFR	71.5	mL/min	Normal

Disease-Specific Risk Probabilities (ML Model)

Disease Domain	Risk (%)	Threshold	Status
Cardiovascular Disease	24.0%	25%	High Risk

CONCLUSION

✅ OmniSensus Medical directly addresses all 5 challenges of Problem Domain 5: Healthcare & Medical Analytics

OmniSensus Medical delivers an end-to-end, production-grade AI-powered clinical decision support platform — from biomarker input to automated PDF report generation.

The composite health scoring system (cardiovascular + metabolic + renal) gives clinicians a single, unified, interpretable wellness score — eliminating fragmented manual analysis.

SHAP-based explainability ensures every AI-generated risk score is transparent and clinically justified — building trust in AI-assisted clinical decisions.

The three-role architecture (Admin, Doctor, Patient) delivers purpose-built, secure, role-isolated workflows reflecting real hospital operational hierarchies.

All 5 Domain 5 challenges are addressed:

- ① Rising disease prevalence → ML early-warning system.
- ② Delayed diagnosis → automated 3-step wizard.
- ③ Resource constraints → admin triage & bed monitoring.
- ④ Data fragmentation → unified 27-table schema.
- ⑤ Predictive uncertainty → ARIMA forecasting + readmission risk model.

Platform demonstrates that production-grade ML + REST API + relational DB + modern web frontend is achievable within an academic timeline.

FUTURE SCOPE

Clinical Expansion

- Additional disease models: Liver Disease (NAFLD), Stroke Risk, Anaemia — each with validated public datasets.
- DICOM medical imaging integration (chest X-ray, ECG) for multi-modal risk combining biomarkers + imaging.
- HL7/FHIR standard compliance for interoperability with existing hospital EMR and HIS systems.

Technical Improvements

- Continuous monthly model retraining pipeline on real platform data with AUC regression gates.
- Federated learning — hospitals train locally without sharing raw patient data, preserving privacy.
- Bayesian uncertainty quantification — report $P(\text{disease}) \pm 95\% \text{ CI}$, not just a point estimate.

AI & LLM Enhancements

- Upgrade Qwen 0.5B → Phi-3 Mini 3.8B on production tier for richer generative PharmaBot responses.
- Natural language vitals input — doctor types "glucose 145, BP 140/90" → LLM extracts structured data.
- LLM-generated clinical narrative summaries per run for integration into patient discharge notes.

Platform & Accessibility

- React Native mobile app — patient health monitoring, medication reminders, PharmaBot on mobile.
- Telemedicine integration — doctor-patient video consultations linked to diagnostic run results.

REFERENCES

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- [9] Kaggle — Pima Indians Diabetes Dataset. Available: <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>
- [10] UCI Machine Learning Repository — CKD Dataset. Available: <https://archive.ics.uci.edu>

 **GitHub:** <https://github.com/parthvadodariya-616/omnisensus>

 **Live:** <https://omnisensus.netlify.app/>

Thank You

OmniSensus Medical — AI-Powered Clinical Intelligence

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